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Bend – Stretch Coupling and Nonlinear Vibrational Dynamics of a Carbon Dioxide Molecule

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Chedin's strongly nonlinear potential is used to describe the vibrational dynamics of a carbon dioxide molecule. Amplitude – phase variables are introduced to separate slow and fast processes. The nonlinear coupling between bend and symmetric stretch vibrations that manifests itself as a slow process is investigated in detail. Efficient intermode energy redistribution is shown to occur, and characteristic frequencies of the energy exchange between the bend and symmetric stretch mode is estimated.

Key words: carbon dioxide, nonlinear intramolecular potential, vibrational energy redistribution, coupled oscillators, intermode energy exchange

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1 Introduction

Carbon dioxide plays an important role in environmental phenomena, industrial and scientific applications for laser systems and as a supercritical solvent in large scale technical extraction processes. Thus, great effort has been spent to understand its properties and behavior on micro- and macroscopic levels.

Energy exchange is essential for most processes concerning laser operation and chemical reactions [1]–[4]. The energy paths of CO_2 lasers involve the energy transitions between all (symmetric and asymmetric stretch and bend) CO_2 modes as well as intermolecular energy exchange. Although energy transfer mechanisms were widely studied experimentally, theoretically many points remain unclear even for dilute systems. At high excitation, carbon dioxide vibrational dynamics is complicated by the nonlinearity of its intramolecular potential energy that is especially important due to strong resonance between the bend and symmetric stretch modes, the vibrational frequency of the latter being approximately twice as high as the former.

The energy redistribution between the symmetric stretch and bend modes can strongly enhance carbon dioxide vibrational energy release to the solvent because there are many indications that the main pathways for vibrational energy relaxation lead via the lowest frequency vibrational modes [5]–[7]. At the same time, the equations describing the dynamics of CO_2 laser emission usually involve terms corresponding to energy flux directly from the symmetric stretch mode to the solvent without any connection to bend mode dynamics [8].

We confine ourselves to a classical mechanical consideration as the starting point of a theoretical treatment of nonlinear dynamics of carbon dioxide. On one hand, this treatment lets us clearly understand the peculiarities of CO_2 intermode interactions. On the other hand, there are many strong correspondences between classical and quantum mechanical treatments of vibrational motion [9, 10]. In

this paper we start to systematically investigate the consequences of the nonlinearity in the CO_2 intra- and intermolecular dynamics.

The paper is organized as follows. The next section is devoted to the description of Chedin's potential [11] and different representations of the equations of motion of a carbon dioxide molecule. In Sec. 3 the slow variables are identified. The dynamical equations for these variables provide evidence for energy redistribution between bend and symmetric stretch modes. The results of a numerical analysis of the slow variables dynamics and the comparison with numerical results that take into account fast motions are analyzed in Sec. 4.

2 CO_2 molecule dynamics based on Chedin's potential

2.1 Chedin's coordinates and potential

CO_2 is a linear molecule with four internal degrees of freedom one of which is a quasirotation. Thus, its ground state potential energy can be represented in terms of three intramolecular coordinates. Chedin's coordinates [11] R_1^*, R_2^* and R_3^* that correspond to the symmetric stretch, bend and asymmetric stretch deformations of the molecule, respectively, are determined by the relations:

$$\begin{aligned} R_1^* &= a_1\left(2 - r_{\rm eq}\frac{r_{21}+r_{23}}{r_{21}r_{23}}\right), \\ R_2^* &= a_2\alpha, \qquad R_3^* = a_3 r_{\rm eq}\frac{r_{21}-r_{23}}{r_{21}r_{23}}, \end{aligned} \tag{1}$$

where

$$\begin{aligned} r_{21} &= \sqrt{(x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2}, \\ r_{23} &= \sqrt{(x_3-x_2)^2+(y_3-y_2)^2+(z_3-z_2)^2} \end{aligned} \tag{2}$$

are the bond lengths between the oxygen (labelled 1 and 3) and carbon (labelled 2) atoms, $\pi-\alpha$ is the angle between these bonds, $r_{\rm eq}$ is the equilibrium bond length (the equilibrium distance between the carbon and oxygen atoms). $\{x_i, y_i, z_i\}$ is the set of ith atom coordinates in the Eckart frame [12, 13]. Chedin's coordinates attain finite values at infinite bond lengths.

Table 1. Chedin's potential constants (in $\rm cm^{-1}$) [11]

F_1	710.07727	F_{202}	-2.920908
F_2	335.47392	F_{022}	-2.7798606
F_3	1231.5194	F_{320}	0.0091365187
F_{300}	-14.182262	F_{302}	0.60846240
F_{400}	-0.25878799	F_{140}	-0.11354617
F_{500}	0.047436019	F_{104}	0.10111166
F_{600}	0.0021151736	F_{420}	-0.084004111
F_{040}	0.82616740	F_{402}	-0.10287321
F_{060}	0.0074460564	F_{240}	-0.00891669000
F_{004}	-1.240548	F_{204}	-0.14007476
F_{006}	-0.011543574	F_{042}	-0.010998372
F_{120}	-29.247842	F_{024}	-0.00090515525
F_{102}	-76.546356	F_{122}	-0.12538577
F_{220}	0.63760223	F_{222}	-0.042680485

For CO_2, the parameter values are $a_1 = 20.294$, $a_2 = 7.665$, $a_3 = 14.248$, $r_{\rm eq} = 1.161$Å .

It is convenient to represent Chedin's potential as a sum of three terms

$$U = U_0 + U_{\rm sma} + U_{\rm cpl} \tag{3}$$

where

$$U_0 = F_1 R_1^{*2} + F_2 R_2^{*2} + F_3 R_3^{*2} \tag{4}$$

is the harmonic part of the potential and

$$\begin{aligned} U_{\rm sma} &= F_{300}R_1^{*3} + F_{400}R_1^{*4} + F_{500}R_1^{*5} \\ &+ F_{600}R_1^{*6} + F_{040}R_2^{*4} + F_{060}R_2^{*6} \\ &+ F_{004}R_3^{*4} + F_{006}R_3^{*6}, \end{aligned} \tag{5}$$

$$\begin{aligned} U_{\rm cpl} &= F_{120}R_1^*R_2^{*2} + F_{102}R_1^*R_3^{*2} + F_{220}R_1^{*2}R_2^{*2} \\ &+ F_{202}R_1^{*2}R_3^{*2} + F_{022}R_2^{*2}R_3^{*2} + F_{320}R_1^{*3}R_2^{*2} \\ &+ F_{302}R_1^{*3}R_3^{*2} + F_{140}R_1^*R_2^{*4} + F_{104}R_1^*R_3^{*4} \\ &+ F_{420}R_1^{*4}R_2^{*2} + F_{402}R_1^{*4}R_3^{*2} + F_{240}R_1^{*2}R_2^{*4} \\ &+ F_{204}R_1^{*2}R_3^{*4} + F_{042}R_2^{*4}R_3^{*2} + F_{024}R_2^{*2}R_3^{*4} \\ &+ F_{122}R_1^*R_2^{*2}R_3^{*2} + F_{222}R_1^{*2}R_2^{*2}R_3^{*2} \end{aligned} \tag{6}$$

stand for the single mode anharmonic and intermode coupling parts of the potential, correspondingly. Constants that appear in Chedin's potential are determined from spectroscopic measurements (Table 1). They are labelled as follows: the first index corresponds to the order of the symmetric stretch variable R_1^*; the second index corresponds to the order of the bend variable R_2^*; and the third index corresponds to the order of the asymmetric stretch variable R_3^*.

Newtonian equations of motion for oxygen and carbon atoms were numerically integrated using molecular dynamics techniques. The coupling between bend and symmetric stretch vibrations due to corresponding terms of Chedin's potential was evident at different initial conditions. This coupling leads to energy exchange between the symmetric stretch and bend vibrational modes that bears periodic features. Depending on the initial conditions, the period of the energy exchange ranges from several to tens of bending mode vibrational periods.

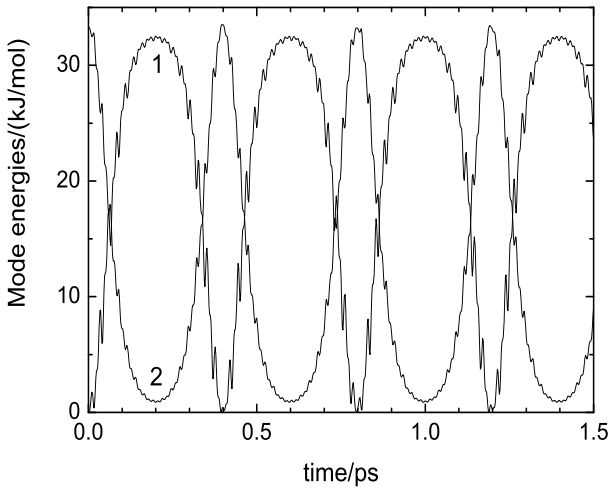


FIG. 1. Molecular dynamics simulation results for the bend – symmetric stretch energy exchange in a CO_2 molecule modelled with the Chedin's potential Eqs. (3) through (6).

Also, the amplitude of energy exchange depends on the initial conditions.

Fig. 1 illustrates the specific character of the intramolecular energy redistribution. Initially, the bend mode is excited to around 9 600 K (1 kJ/mol corresponds to approximately 120.3 K or 83.6 cm^{-1} for one vibrational degree of freedom). Due to the intermode coupling the symmetric stretch becomes excited as well while the asymmetric stretch mode remains unexcited. At given initial conditions, the period of energy exchange is around 400 fs, which can be compared with bend and symmetric stretch vibrational periods of 49.6 and 24.7 fs, respectively.

Below we simplify the model to separate the main terms of the potential energy responsible for this coupling.

2.2 Equations of motion in Chedin's coordinates

The kinetic energy of the system under consideration can be written as follows:

$$K = \frac{m_O}{4}(\dot{q}_1^2 + \dot{q}_2^2 + \dot{q}_3^2 + q_2^2\dot{\varphi}^2), \qquad (7)$$

where q_1 and q_3 are the normal coordinates for symmetric and asymmetric stretches, q_2 and φ are the normal coordinates for doubly degenerate bend, respectively. The normal coordinates are expressed in terms of Eckart's coordinates of CO_2 atoms by the following formulae

$$\begin{aligned} q_1 &= x_3 - x_1 - 2r_{eq}, \quad q_3 = \gamma(x_3 + x_1), \\ q_2 &= 2\gamma\sqrt{y_3^2 + z_3^2}, \quad \varphi = \arctan\frac{y_3}{z_3} \end{aligned} \qquad (8)$$

where

$$\gamma = \sqrt{1 + 2m_O/m_C} \qquad (9)$$

and m_C and m_O are the masses of carbon and oxygen atoms, respectively.

To write down the equations of motion it is desirable to express the normal coordinates (8) in terms of Chedin's coordinates (1). Accomplishing this we

take into consideration the relations

$$\begin{aligned} r_{21}^2 &= \frac{1}{4}\left[(q_1 + 2r_{\rm eq} - \gamma q_3)^2 + \gamma^2 q_2^2\right], \\ r_{23}^2 &= \frac{1}{4}\left[(q_1 + 2r_{\rm eq} + \gamma q_3)^2 + \gamma^2 q_2^2\right] \end{aligned} \quad (10)$$

that connect normal and Eckart's coordinates according to Eqs. (1), (2) and (8).

It is worth to investigate nonlinear effects caused exclusively by the Chedin's potential nonlinearity. In this case the kinematic nonlinearities contained in Eqs. (10) can be neglected and the expressions for the distances r_{21} and r_{23} assume the linear form

$$\begin{aligned} r_{21} &= r_{\rm eq} + (q_1 - \gamma q_3)/2, \\ r_{23} &= r_{\rm eq} + (q_1 + \gamma q_3)/2. \end{aligned} \quad (11)$$

For the following consideration it is convenient to introduce new coordinates

$$R_1 = a_1^{-1} R_1^*, \quad R_2 = a_2^{-1} R_2^*, \quad R_3 = a_3^{-1} R_3^*, \quad (12)$$

to which we also refer as Chedin's ones. The latter in linear approximation

$$R_1 = -\frac{q_1}{r_{\rm eq}}, \quad R_2 = \gamma \frac{q_2}{r_{\rm eq}}, \quad R_3 = -\gamma \frac{q_3}{r_{\rm eq}} \quad (13)$$

allow one to represent the kinetic energy of the system under consideration in terms the time derivatives of Chedin's coordinates

$$K = \frac{1}{2}(\mu_1 \dot{R}_1^2 + \mu_2 \dot{R}_2^2 + \mu_3 \dot{R}_3^2 + \mu_1 R_1^2 \dot{\varphi}^2), \quad (14)$$

where inertia coefficients

$$\mu_1 = \frac{m_O r_{\rm eq}^2}{2}, \quad \mu_2 = \mu_3 = \frac{m_O r_{\rm eq}^2}{2\gamma^2} \quad (15)$$

as can be easily derived by comparing expressions (7) and (14) and taking into account the linearized relations (13).

The carbon dioxide model as defined by Eqs. (1) through (6) and (14) takes into account the linear kinematic interaction between stretch and bend *local* modes that was earlier investigated for water- and benzene-like molecules (see Refs. [14]-[16] and references therein). The corresponding terms can be singled out from the kinetic energy Eq. (14) after the transition to local coordinates. These terms give linear contributions to the equations of motion and lead to beats in the *local* coordinates time evolution because the local coordinates are not normal ones.

Of course, the equations of motion contain kinematic nonlinear terms originating from the kinetic energy as well. This nonlinearity is shown to be important for the stretch – bend coupling, e. g. for deuterated water molecules [17]. However, its consideration is beyond the scope of the current paper. Furthermore, only in-plane motion will be considered here. It means we neglect the quasirotation (the last term in Eq. (14) for kinetic energy).

The expressions for the kinetic (14) and potential (3) energies are used to construct the equations of motion for CO_2 in terms of Chedin's coordinates (index j runs from 1 through 3):

$$\ddot{R}_j + \omega_j^2 R_j = \omega_j^2 f_j, \quad (16)$$

where

$$\omega_j = 2\pi\nu_j = a_j \sqrt{\frac{2F_j}{\mu_j}} \quad (17)$$

correspond to the harmonic frequencies of symmetric stretch ($\nu_1 = 1352{\rm cm}^{-1}$), bend ($\nu_2 = 672{\rm cm}^{-1}$) and asymmetric stretch ($\nu_3 = 2393{\rm cm}^{-1}$), and

$$f_j = -\frac{1}{2F_j a_j^2} \frac{\partial}{\partial R_j}(U_{\rm sma} + U_{\rm cpl}) \quad (18)$$

are dimensionless forces. In accordance with Eq. (3)–(6) these forces are functions of Chedin's coordinates. It is convenient to represent these dimensionless forces in terms of the anharmonic single mode ($f_j^{\rm sma}$) and coupling ($f_j^{\rm cpl}$) contributions:

$$\begin{aligned} f_1^{\rm sma} &= k_{200}^{(1)} R_1^2 + k_{300}^{(1)} R_1^3 - k_{400}^{(1)} R_1^4 - k_{500}^{(1)} R_1^5, \\ f_2^{\rm sma} &= -k_{030}^{(2)} R_2^3 - k_{050}^{(2)} R_2^5, \\ f_3^{\rm sma} &= k_{003}^{(3)} R_3^3 + k_{005}^{(3)} R_3^5, \end{aligned} \quad (19)$$

$$
\begin{aligned}
f_1^{\rm cpl} &= k_{020}^{(1)}R_2^2 - k_{120}^{(1)}R_1R_2^2 - k_{220}^{(1)}R_1^2R_2^2 \\
&+ k_{040}^{(1)}R_2^4 + k_{320}^{(1)}R_1^3R_2^2 + k_{140}^{(1)}R_1R_2^4 \\
&+ k_{002}^{(1)}R_3^2 + k_{102}^{(1)}R_1R_3^2 - k_{202}^{(1)}R_1^2R_3^2 \\
&+ k_{022}^{(1)}R_2^2R_3^2 - k_{004}^{(1)}R_3^4 + k_{302}^{(1)}R_1^3R_3^2 \\
&+ k_{104}^{(1)}R_1R_3^4 + k_{122}^{(1)}R_1R_2^2R_3^2, \\
f_2^{\rm cpl} &= k_{110}^{(2)}R_1R_2 - k_{210}^{(2)}R_1^2R_2 - k_{310}^{(2)}R_1^3R_2 \\
&+ k_{130}^{(2)}R_1R_2^3 + k_{410}^{(2)}R_1^4R_2 + k_{230}^{(2)}R_1^2R_2^3 \\
&+ k_{012}^{(2)}R_2R_3^2 + k_{032}^{(2)}R_2^3R_3^2 + k_{014}^{(2)}R_1R_3^4 \\
&+ k_{112}^{(2)}R_1R_2R_3^2 + k_{212}^{(2)}R_1^2R_2R_3^2, \\
f_3^{\rm cpl} &= k_{101}^{(3)}R_1R_3 + k_{201}^{(3)}R_1^2R_3 + k_{021}^{(3)}R_2^2R_3 \\
&- k_{301}^{(3)}R_1^3R_3 + k_{041}^{(3)}R_2^4R_3 - k_{103}^{(3)}R_1R_3^3 \\
&+ k_{401}^{(3)}R_1^4R_3 + k_{023}^{(3)}R_2^2R_3^3 + k_{203}^{(3)}R_1^2R_3^3 \\
&+ k_{121}^{(3)}R_1R_2^2R_3 + k_{221}^{(3)}R_1^2R_2^2R_3. \qquad (20)
\end{aligned}
$$

Dimensionless force constants $k_{ijk}^{(\alpha)}$ are defined as positive coefficients of the coordinates products in accordance with Eqs. (5), (6) and (18). Numerical values of k coefficients are listed in Table 2.

The subscripts of the force constants $k_{ijk}^{(\alpha)}$ indicate the order of the symmetric stretch, bend and asymmetric stretch mode variables of the corresponding term while the superscripts denote the modes for which the corresponding forces are written. Among all the k coefficients only 25 are independent because they are determined Chedin's potential constants F:

$$k_{\{i_\alpha-1\}}^{(\alpha)} = i_\alpha a_1^{i_1}a_2^{i_2}a_3^{i_3}|F_{i_1i_2i_3}|/(2a_\alpha^2F_\alpha), \qquad (21)$$

where $\{i_\alpha - 1\}$ means that all the subscripts of k are the same as of F except of i_α that is replaced by $i_\alpha - 1$. For example,

$$F_2a_2^2k_{110}^{(2)} = 2F_1a_1^2k_{020}^{(1)} = -a_1a_2^2F_{120}. \qquad (22)$$

2.3 Equations of motion for amplitude and phase variables

It is convenient to study the motion of the system in terms of amplitude and phase variables. To introduce these quantities we rewrite the equations of motion (16) as the system of first order differential equations

$$\dot{R}_j = \omega_jP_j, \quad \dot{P}_j = -\omega_jR_j + \omega_jf_j. \qquad (23)$$

Then, introducing the complex variables [18]

$$c_j = R_j + iP_j, \quad c_j^* = R_j - iP_j, \qquad (24)$$

and adding the first equation in (23) to the second one multiplied by $i = \sqrt{-1}$ the equations

$$\dot{c}_j = -i\omega_jc_j + i\omega_jf_j. \qquad (25)$$

for complex variables are obtained.

Table 2. Values of dimensionless force constants $k_{ijk}^{(\alpha)}$

$k_{200}^{(1)}$	0.6080	$k_{012}^{(2)}$	1.682
$k_{300}^{(1)}$	0.3002	$k_{310}^{(2)}$	0.2276
$k_{400}^{(1)}$	1.396	$k_{130}^{(2)}$	0.8071
$k_{500}^{(1)}$	1.516	$k_{112}^{(2)}$	1.540
$k_{020}^{(1)}$	0.05962	$k_{410}^{(2)}$	42.47
$k_{002}^{(1)}$	0.5392	$k_{230}^{(2)}$	1.286
$k_{120}^{(1)}$	0.05276	$k_{212}^{(2)}$	10.64
$k_{102}^{(1)}$	0.8351	$k_{032}^{(2)}$	0.7821
$k_{220}^{(1)}$	0.02301	$k_{014}^{(2)}$	0.1112
$k_{202}^{(1)}$	5.295	$k_{003}^{(3)}$	0.4090
$k_{022}^{(1)}$	0.05189	$k_{005}^{(3)}$	1.159
$k_{040}^{(1)}$	0.01360	$k_{101}^{(3)}$	1.261
$k_{004}^{(1)}$	0.1446	$k_{201}^{(3)}$	0.9768
$k_{320}^{(1)}$	5.725	$k_{021}^{(3)}$	0.1326
$k_{302}^{(1)}$	24.23	$k_{301}^{(3)}$	4.129
$k_{140}^{(1)}$	0.04335	$k_{121}^{(3)}$	0.1214
$k_{104}^{(1)}$	8.130	$k_{103}^{(3)}$	0.6765
$k_{122}^{(1)}$	0.7169	$k_{401}^{(3)}$	14.17
$k_{030}^{(2)}$	0.2894	$k_{221}^{(3)}$	0.8386
$k_{050}^{(2)}$	0.2298	$k_{203}^{(3)}$	19.02
$k_{110}^{(2)}$	1.769	$k_{041}^{(3)}$	0.03083
$k_{210}^{(2)}$	0.7828	$k_{023}^{(3)}$	0.01753

Switching to the exponential representation of complex variables

$$c_j = A_j e^{-i\varphi_j}, \quad c_j^* = A_j e^{i\varphi_j}, \tag{26}$$

where A_j and φ_j are amplitude and phase variables, respectively, after separation of real and imaginary parts, equations (25) take the form

$$\begin{aligned} A_j\dot{\varphi}_j &= -i\omega_j A_j - \mathcal{I}\mathrm{m}\{\omega_j f_j\}, \\ \dot{A}_j &= \mathcal{R}\mathrm{e}\{\omega_j f_j\}. \end{aligned} \tag{27}$$

Keeping in mind that the dimensionless forces are real and using the Euler's formula

$$e^{i\varphi} = \cos\varphi + i\sin\varphi$$

equations (27) can be rewritten as follows:

$$\begin{aligned} A_j\dot{\varphi}_j &= -\omega_j A_j - \omega_j f_j \cos\varphi_j, \\ \dot{A}_j &= -\omega_j f_j \sin\varphi_j. \end{aligned} \tag{28}$$

3 Dynamics of slow variables

3.1 Dynamical equations

The values of the constants listed in Table 1 make it possible to simplify the problem under consideration. For not too strong excitations ($R_j \sim 0.1$) we can consider the largest contributions only in the equations of motion (16). Then the potential energy and the dimensionless forces read

$$\begin{aligned} U &= U_0 + F_{300}a_1^3R_1^3 \\ &+ F_{120}a_1a_2^2R_1R_2^2 + F_{102}a_1a_3^2R_1R_3^2, \end{aligned} \tag{29}$$

$$\begin{aligned} f_1 &= k_{200}^{(1)}R_1^2 + k_{020}^{(1)}R_2^2 + k_{002}^{(1)}R_3^2, \\ f_2 &= k_{110}^{(2)}R_1R_2, \\ f_3 &= k_{101}^{(3)}R_1R_3. \end{aligned} \tag{30}$$

Therefore, Eqs. (16) take the form

$$\begin{aligned} \ddot{R}_1 + \omega_1^2R_1 &= \omega_1^2(k_1'R_1^2 + k_1R_2^2 + k_1''R_3^2), \\ \ddot{R}_2 + \omega_2^2R_2 &= \omega_2^2k_2R_1R_2, \\ \ddot{R}_3 + \omega_3^2R_3 &= \omega_3^2k_3R_1R_3, \end{aligned} \tag{31}$$

where simplified notations

$$\begin{aligned} &k_1' \equiv k_{200}^{(1)}, \quad k_1 \equiv k_{020}^{(1)}, \quad k_1'' \equiv k_{002}^{(1)}, \\ &k_2 \equiv k_{110}^{(2)}, \quad k_3 \equiv k_{101}^{(3)} \end{aligned} \tag{32}$$

are introduced.

Eqs. (31) can be rewritten in terms of complex variables (24):

$$\begin{aligned} \dot{c}_1 &= -i\omega_1c_1 + \frac{i\omega_1}{4}[k_1(c_2 + c_2^*)^2 \\ &+ k_1'(c_1 + c_1^*)^2 + k_1''(c_3 + c_3^*)^2], \\ \dot{c}_2 &= -i\omega_2c_2 + \frac{i\omega_2k_2}{4}(c_2 + c_2^*)(c_1 + c_1^*), \\ \dot{c}_3 &= -i\omega_3c_3 + \frac{i\omega_3k_3}{4}(c_3 + c_3^*)(c_1 + c_1^*) \end{aligned} \tag{33}$$

or using Eq. (28) in terms of amplitudes and phases

$$\begin{aligned} \dot{A}_1 &= -\omega_1\sin\varphi_1(k_1A_2^2\cos^2\varphi_2 \\ &+ k_1'A_1^2\cos^2\varphi_1 + k_1''A_3^2\cos^2\varphi_3), \\ A_1\dot{\varphi}_1 &= \omega_1A_1 - \omega_1\cos\varphi_1(k_1A_2^2\cos^2\varphi_2 \\ &+ k_1'A_1^2\cos^2\varphi_1 + k_1''A_3^2\cos^2\varphi_3), \\ \dot{A}_2 &= -\frac{\omega_2k_2}{2}A_1A_2\cos\varphi_1\sin^2\varphi_2, \\ \dot{\varphi}_2 &= \omega_2 - \omega_2k_2A_1\cos\varphi_1\cos^2\varphi_2, \\ \dot{A}_3 &= -\frac{\omega_3k_3}{2}A_1A_3\cos\varphi_1\sin^2\varphi_3, \\ \dot{\varphi}_3 &= \omega_3 - \omega_3k_3A_1\cos\varphi_1\cos^2\varphi_3. \end{aligned} \tag{34}$$

The mode energies in the harmonic approximation consist of kinetic and potential contributions

$$\begin{aligned} E_j &= \frac{\mu_j}{2}\dot{R}_j^2 + F_ja_j^2R_j^2 \\ &= \frac{\mu_j}{2}(\dot{R}_j^2 + \omega_j^2R_j^2), \end{aligned} \tag{35}$$

where Eq. (17) is used. Taking into account the complex [Eq. (24)] and amplitude [Eq. (26)] representations of dynamic variables, the mode energies Eq. (35) can be rewritten as

$$E_j = \frac{\mu_j\omega_j^2}{2}c_jc_j^* = \frac{\mu_j\omega_j^2}{2}A_j^2. \tag{36}$$

The products and powers of trigonometric functions can be easily removed from Eqs. (34)

$$\begin{aligned}
\dot{A}_1 &= -\frac{\omega_1}{4}[(2k_1A_2^2 + k_1'A_1^2 + 2k_1''A_3^2)\sin\varphi_1 \\
&+ k_1A_2^2(\sin(\varphi_1+2\varphi_2)+\sin(\varphi_1-2\varphi_2)) \\
&+ k_1''A_3^2(\sin(\varphi_1+2\varphi_3)+\sin(\varphi_1-2\varphi_3)) \\
&+ k_1'A_1^2\sin 3\varphi_1], \\
A_1\dot{\varphi}_1 &= A_1\omega_1 \\
&- \frac{\omega_1}{4}[(2k_1A_2^2 + 3k_1'A_1^2 + 2k_1''A_3^2)\cos\varphi_1 \\
&+ k_1A_2^2(\cos(\varphi_1+2\varphi_2)+\cos(\varphi_1-2\varphi_2)) \\
&+ k_1''A_3^2(\cos(\varphi_1+2\varphi_3)+\cos(\varphi_1-2\varphi_3)) \\
&+ k_1'A_1^2\cos 3\varphi_1], \\
\dot{A}_2 &= \frac{\omega_2k_2}{4}A_1A_2 \\
&\times [\sin(\varphi_1-2\varphi_2)-\sin(\varphi_1+2\varphi_2)], \\
A_2\dot{\varphi}_2 &= A_2\omega_2 - \frac{\omega_2k_2}{4}A_1A_2[2\cos\varphi_1 \\
&+ \cos(\varphi_1+2\varphi_2)+\cos(\varphi_1-2\varphi_2)], \\
\dot{A}_3 &= \frac{\omega_3k_3}{4}A_1A_3 \\
&\times [\sin(\varphi_1-2\varphi_3)-\sin(\varphi_1+2\varphi_3)], \\
A_3\dot{\varphi}_3 &= A_3\omega_3 - \frac{\omega_3k_3}{4}A_1A_3[2\cos\varphi_1 \\
&+ \cos(\varphi_1+2\varphi_3)+\cos(\varphi_1-2\varphi_3)], \qquad (37)
\end{aligned}$$

where the r.h.s. terms show explicitly the phase combinations that determine the time evolution of the amplitudes and phases. As the ratio of the harmonic frequencies of symmetric stretch and bend vibrations is approximately equal to 2, the strong resonance between these vibrations allows to separate the fast and slow terms in the equations of motion. The slow terms contain the difference of phases

$$\Phi = \varphi_1 - 2\varphi_2. \qquad (38)$$

Thus, it is possible to average Eqs. (37) over the bend vibrational period $2\pi/\omega_2$ to remove all the terms that contain fast phases φ_1, $3\varphi_1$, $(\varphi_1+2\varphi_2)$ and $(\varphi_1 \pm 2\varphi_3)$. Then Eqs. (37) take the form

$$\begin{aligned}
\dot{A}_1 &= -\frac{\omega_1k_1}{4}A_2^2\sin\Phi, \qquad (39) \\
\dot{\varphi}_1 &= \omega_1 - \frac{\omega_1k_1}{4A_1}A_2^2\cos\Phi, \\
\dot{A}_2 &= \frac{\omega_2k_2}{4}A_1A_2\sin\Phi, \qquad (40) \\
\dot{\varphi}_2 &= \omega_2 - \frac{\omega_2k_2}{4}A_1\cos\Phi, \\
\dot{A}_3 &= 0, \qquad (41) \\
\dot{\varphi}_3 &= \omega_3.
\end{aligned}$$

For smoothed amplitudes and phases we preserve the same notation as in Eqs. (37).

Eqs. (41) show that in this approximation the asymmetric mode is harmonic and independent from the others. The problem is reduced to the treatment of coupling solely between the symmetric stretch and bend modes.

Eliminating $\sin\Phi$ from the first lines of Eqs. (39), (40) and integrating the resulting expression over time one obtains the relationship between the smoothed amplitudes

$$A_1^2 + kA_2^2 = e_0 \qquad (42)$$

with dimensionless coefficient

$$k = \frac{\omega_1k_1}{\omega_2k_2}. \qquad (43)$$

For the model under consideration $k \approx 0.0678$.

The integration constant

$$e_0 = A_{10}^2 + kA_{20}^2, \qquad (44)$$

is represented through initial values of the amplitudes A_{10} and A_{20}. This constant is related to the initial mode harmonic energies E_{10} and E_{20} defined in accordance with Eq. (36) as

$$\frac{\mu_1\omega_1^2}{2}e_0 = E_{10} + \frac{\omega_1}{2\omega_2}E_{20}, \qquad (45)$$

where Eq. (22) rewritten in the form

$$2\mu_1\omega_1^2k_1 = \mu_2\omega_2^2k_2 \qquad (46)$$

was used in the last term.

This equation can be given a dimensionless form by dividing it by $\mu_1\omega_1^2/2$

$$e_0 = e_{10} + \frac{\omega_1}{2\omega_2} e_{20}. \tag{47}$$

It means that all energies are determined in units of $\mu_1\omega_1^2/2 = a_1^2 F_1$ (F_1 =710 cm^{-1} or 8.49 kJ/mol) and for arbitrary time

$$e_1 = A_1^2, \quad e_2 = k(2\omega_2/\omega_1)A_2^2.$$

Indeed, Eq. (42) reflects the energy conservation for the smoothed variables because the coefficient of e_{20} in Eq. (47) is close to 1 (it is approximately equal to 1.006). This coefficient is determined by harmonic characteristics of the system and becomes equal to one for the ideal resonance condition $\omega_1 = 2\omega_2$ although the excess energy ($e_0 > e_{10} + e_{20}$) is due to the nonlinear terms.

Eq. (42) can be used for simplifying the description to the level of a single degree of freedom

$$\begin{aligned} \dot{A}_1 &= -\frac{\omega_2 k_2}{4}(e_0 - A_1^2)\sin\Phi, \\ \dot{\Phi} &= \omega_1 - 2\omega_2 - \frac{\omega_2 k_2}{4A_1}(e_0 - 3A_1^2)\cos\Phi. \end{aligned} \tag{48}$$

However, these equations cannot be considered as the equations of motion for the symmetric stretch mode because of Eq. (38). The slow angular velocity $\dot{\Phi}$ contains small constant dephasing contribution $(\omega_1 - 2\omega_2)$ caused by not ideal resonance conditions. The magnitude and the sign of the rate of amplitude change $\dot{A}_1$ strongly depends on the phase difference Φ.

As Eqs. (48) are strongly nonlinear, the behavior of the amplitude A_1 and the difference of phases Φ obviously depend on the initial conditions: the initial energy, its distribution among the modes (A_{10}/A_{20}), and the initial phase difference (Φ_0). After Eqs. (48) are integrated the time dependence of the bending mode amplitude A_2 and the smoothed phases φ_1 and φ_2 can be determined by Eqs. (42),(39) and (40).

3.2 Intermode energy redistribution

The rate of mode energy change can be calculated by multiplying Eqs. (31) with $\mu_j \dot{R}_j$

$$\begin{aligned} \dot{E}_1 &= \mu_1\omega_1^2 k_1 R_2^2 \dot{R}_1, \\ \dot{E}_2 &= 2\mu_2\omega_2^2 k_2 R_1 R_2 \dot{R}_2. \end{aligned} \tag{49}$$

It is convenient to rewrite these equations in amplitude and phase variables and average them over the bending mode vibrational period that preserves only the slow phase Φ in the resulting expressions

$$\begin{aligned} \dot{E}_1 &= \frac{1}{4}\mu_1\omega_1^3 k_1 A_2^2 A_1 \sin\Phi, \\ \dot{E}_2 &= -2\mu_2\omega_2^3 k_2 A_1 A_2^2 \sin\Phi. \end{aligned} \tag{50}$$

Here we used that $\dot{A}_j << \omega_j A_j$ because the smoothed amplitudes are slow variables. In dimensionless form these equations can be written as

$$\begin{aligned} \dot{e}_1 &= \frac{k_1}{2}\omega_1 A_2^2 A_1 \sin\Phi, \\ \dot{e}_2 &= -k_1\omega_2 A_1 A_2^2 \sin\Phi. \end{aligned} \tag{51}$$

Again, Eq. (46) was used to derive the last expression.

The mode energy derivatives are of opposite sign and proportional to the symmetric stretch amplitude and to the square of the bending mode amplitude. Thus, the intermode energy exchange becomes very slow when the bending mode amplitude attains small values.

The absolute values of the r.h.s. terms of Eqs. (51) are not equal to each other and the harmonic part of the energy is not conserved due to intermode coupling term. The rate of its change

$$\dot{e}_1 + \dot{e}_2 = \frac{k_1}{2}(\omega_1 - 2\omega_2) A_1 A_2^2 \sin\Phi. \tag{52}$$

is proportional to the bare dephasing difference $(\omega_1 - 2\omega_2)$ and changes its sign with $\sin\Phi$.

4 Numerical results and analysis

The equations of motion (34) were integrated by means of MATHEMATICA 4.1 for the symmetric

stretch and bending mode motions, where the terms connecting these modes with the asymmetric stretch were omitted ($A_3 = 0, \varphi_3 = 0$). On the other hand, the simplified one degree of freedom equations (48) were integrated as well.

The dynamics represented in Fig. 2 is qualitatively similar to the dynamics shown in Fig. 1. The latter shows the result of MD simulations for the carbon dioxide model based on Chedin's potential including the kinematic nonlinearities, while only potential energy nonlinearities are included in Fig. 2. The initial conditions for both cases are approximately the same. The bending mode was excited to an energy that corresponds to two quanta of the symmetric stretch mode (∼32.36 kJ/mol) at unexcited stretch mode and zero phase difference ($\Phi_0 = 0$). The intermode energy exchange is perfectly periodic with a period $T_E = 740$ fs (Fig. 2) that is slightly longer than in Fig. 1. Obviously, this difference is due to the kinematic nonlinearities not included in Eqs. (34).

The diversity of carbon dioxide dynamics due to different initial conditions is illustrated in Figs. 3 through 6.

Shown in Fig. 3 is the result of the initially high symmetric stretch excitation $E_{10}/E_{20} = 10^2$, again at $\Phi_0 = 0$. The energy exchange period becomes considerably longer and can be made as long as desirable by increasing the initial energy ratio.

The rate of the energy exchange is linear in the stretch amplitude and proportional to the squared bending mode amplitude [see Eqs. (50) and (51)]. It means that the energy exchange is very weak when the bending mode is weakly excited and this state of the system can be maintained for a long time. It explains the growth of the energy exchange period when the ratio E_{10}/E_{20} is quite big.

In the opposite case of equally excited modes $E_{10}/E_{20} = 1$ the "mirror" mode behavior (Fig. 4 for $\Phi_0 = 0$ and Fig. 5 for $\Phi_0 = \pi$) is determined by the energy conservation. The opposite initial phases result in different amplitudes and periods (0.56 and

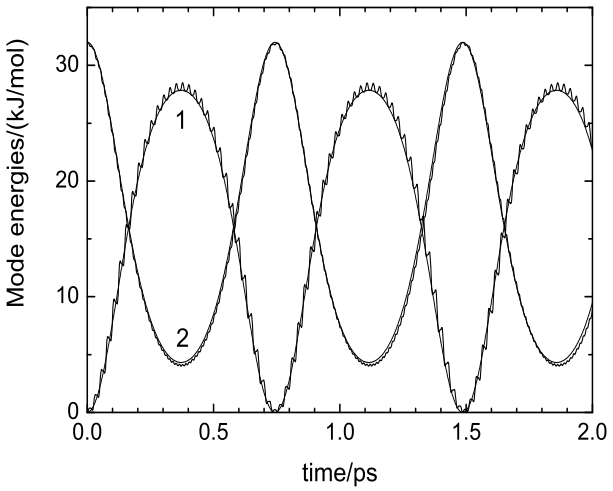


FIG. 2. Bend – symmetric stretch energy exchange at $E_{20} = 32.36$ kJ/mol, unexcited stretch mode and zero phase difference $\Phi_0 = 0$. The symmetric stretch (1) and bend (2) mode energies are shown. The thick smooth curves correspond to the one degree of freedom Eqs.(48), while the thin curves with high frequency details are solutions of the Eqs. (35) through (37) at $A_3 = 0$.

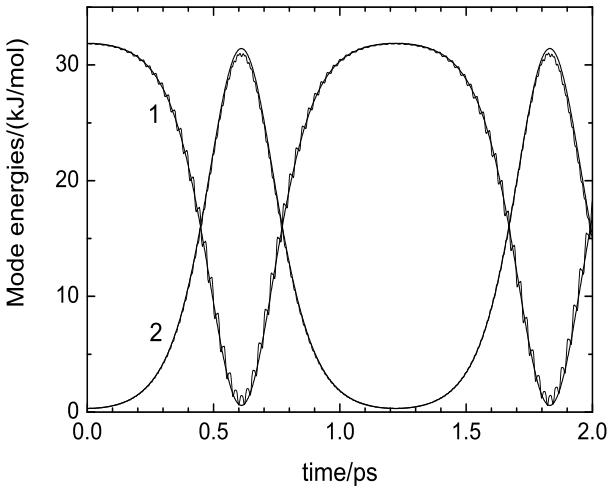


FIG. 3. Bend – symmetric stretch energy exchange at $E_0 = 32.36$ kJ/mol, $E_{10}/E_{20} = 10^2$ and $\Phi_0 = 0$. The line notations are the same as in Fig. 2.

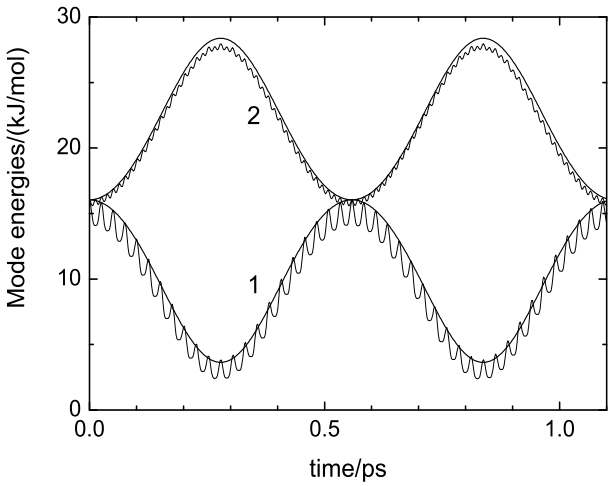


FIG. 4. Bend – symmetric stretch energy exchange at $E_0 = 32.36$ kJ/mol, $E_{10} = E_{20}$ and $\Phi_0 = 0$. The line notations are the same as in Fig. 2.

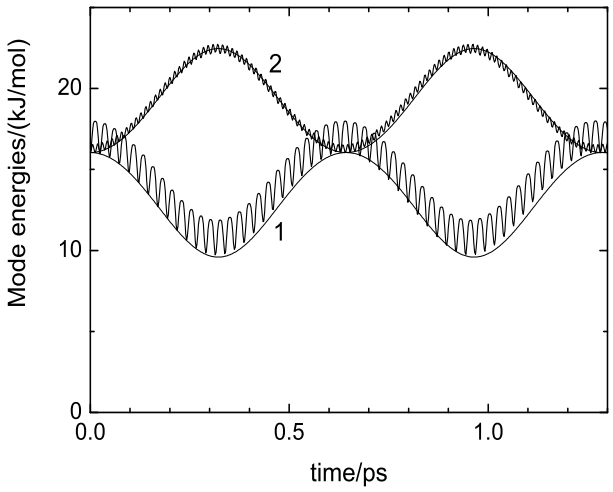


FIG. 5. Bend – symmetric stretch energy exchange at $E_0 = 32.36$ kJ/mol, $E_{10} = E_{20}$ and $\Phi_0 = \pi$. The line notations are the same as in Fig. 2.

0.64 ps, respectively) of the energy exchange.

The results shown in Figs. 2 through 5 clearly indicate that the one degree of freedom equations (48) are a good representation of the slow processes: the smooth curves representing the evolution of the slow variables follow the solutions of the full nonlinear problem. Of course, the latter include high frequency oscillations due to the nonlinear terms with fast phases. It is interesting to note that the high frequency energy deviations in Figs. 4 and 5 are of opposite sign with respect to the smooth curves depicting the slow variables.

Not only the energy exchange period depends strongly on the initial conditions. The initial conditions influence the amplitude of the energy exchange. If initially the mode energy ratio E_{10}/E_{20} is sufficiently big, the intermode energy exchange is almost complete (see Fig. 3) and the phase difference Φ shows the rotating behavior. When $E_{10}/E_{20} = 6.44$ and $\Phi_0 = 0$ the transition to oscillating behavior occurs. The magnitude of the energy exchange becomes asymmetric between two half periods (see Fig. 4) and this asymmetry grows when the ratio E_{10}/E_{20} decreases further. Figs. 4 and 5 illustrate the energy exchange at $E_{10}/E_{20} = 1$.

There are two conditions when the energy exchange between the symmetric stretch and bend motions consists only of small high frequency variations. They are at $E_{10}/E_{20} = 0.395$ for $\Phi_0 = 0$ (see Fig. 6) and $E_{10}/E_{20} = 0.654$ for $\Phi_0 = \pi$ (not shown). These values correspond to the slow variables solutions. For the full nonlinear equations the energy ratios that lead to the minimal intermode energy exchange are slightly different: $E_{10}/E_{20} = 0.405$ and 0.661 for $\Phi_0 = 0$ and π, correspondingly.

Further decrease of the energy ratio does not create physically new trajectories because for the same trajectory the oscillating phase behavior leads to two different energy ratios below and above the point of the absence of the energy exchange. E. g. $E_{10}/E_{20} \simeq 0$ and 6.44 at $\Phi_0 = 0$ are characteristic for the conditions of Fig. 2.

Peculiarities of the phase difference behavior are of interest as well because of the role it plays in the intermode energy flux magnitude and direction.

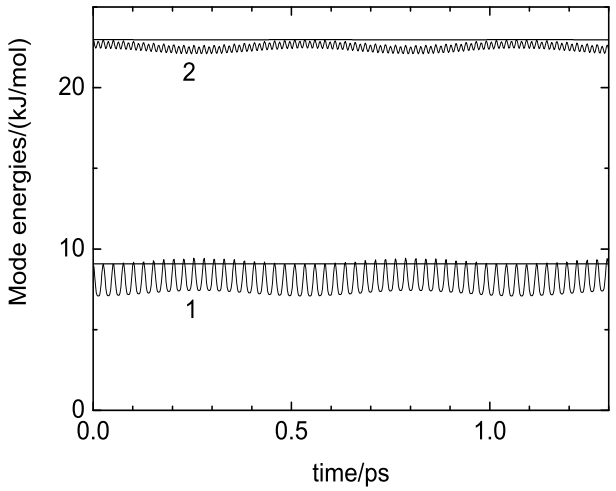


FIG. 6. Bend – symmetric stretch energy exchange at $E_0 = 32.36$ kJ/mol, $E_{10}/E_{20} = 0.395$ and $\Phi_0 = 0$. The line notations are the same as in Fig. 2.

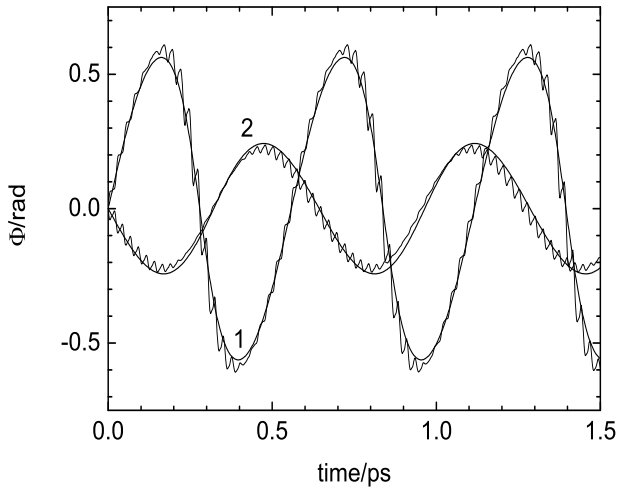


FIG. 7. Oscillating behavior of the phase difference $\Phi = \varphi_1 - 2\varphi_2$ at equal initial energies $E_{10} = E_{20} = 16.18$ kJ/mol, and $\Phi_0 = 0$ (curves 1) and $\Phi_0 = \pi$ (curves 2). For the latter case the difference $\Phi - \pi$ is shown. The meaning of lines is the same as in Fig. 2.

The second of Eqs. (48) implies the fast phase dynamics when the amplitude (and the energy) of the symmetric stretch mode attain small values. Just this situation can be observed for the initial conditions of Fig. 2. This is the case when the transition from the oscillating to rotating phase behavior occurs. At these conditions the phase behavior is unstable and at particular moments of time a sudden jump by π of the slow phase can occur that results in deviations between the solutions of Eqs. (34) and (48).

At other initial conditions the solutions of these equations well correspond to each other. The oscillating phase behavior appears at conditions of Figs. 4 and 5 (see Figs. 7). In the latter case the phase difference Φ is close to π, which was subtracted to put the curves in the figure frame. The rotating phase behavior is illustrated in Fig. 8 corresponding to the initial conditions of Fig. 3. As expected, Φ increases rapidly at $t \simeq 0.6$ and 1.8 ps when the symmetric stretch mode energy is very small, and less rapidly at $t \simeq 1.2$ ps when this energy and the difference $3A_1^2 - e_0$ are big (see Eq. (48)). In both cases $\cos\Phi$ is also not too small. Contrary to this the almost horizontal segments on the angle Φ versus time dependence correspond to $\Phi \simeq \pi(1/2 + k)$ and $\cos\Phi \simeq 0$ (k is an integer).

Fig. 9 illustrates the beating behavior of Chedin's coordinates in the case corresponding to the conditions of Fig. 2. The high frequency vibrations are superimposed on the slow variations of the amplitudes. The difference of phases can hardly be extracted from this figure. However, it is evident that at $t \simeq 0$, 0.75 and 1.5 ps this difference can be easily changed because of low energy content of the symmetric stretch mode. Qualitatively the phase difference behavior is similar to shown in Fig. 8 although it is not stable as discussed above.

5 Conclusions

The contribution of the nonlinear terms of Chedin's potential to the dynamics of a carbon dioxide molecule is investigated. Due to a strong resonance

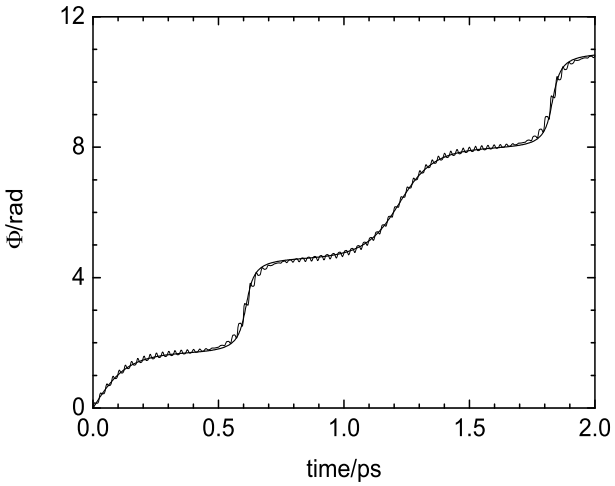


FIG. 8. Rotating behavior of the phase difference $\Phi = \varphi_1 - 2\varphi_2$ at $E_{10}/E_{20} = 10^2$ and $\Phi_0 = 0$.

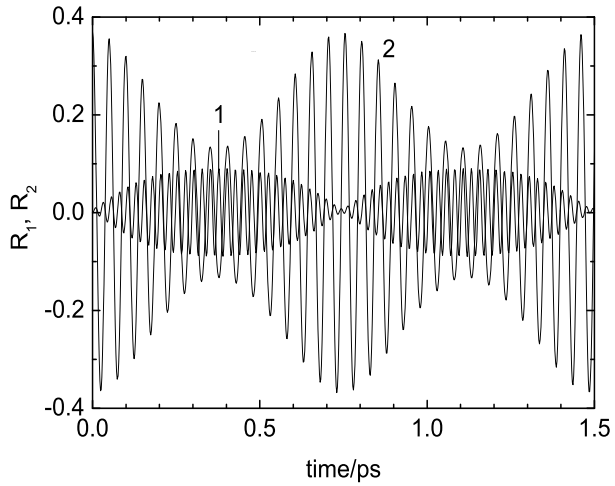


FIG. 9. The beating behavior of the Chedin's coordinates. The initial conditions and the curve notations are the same as in Fig. 2.

between the bend and symmetric stretch mode efficient energy exchange between these modes exists and can be described in terms of slow variables, i. e. the amplitude (or energy) of the symmetric stretch (or, alternatively, bending) mode and the phase difference between them. It is shown that the slow variables dynamics is well separated from the asymmetric stretch mode.

After transition to slow variables the equations of motion of the bend and symmetric stretch modes are reduced to the form of a single degree of freedom that allows us to analyze the carbon dioxide classical dynamics in a great detail. The comparison with solutions of the full dynamical equations indicates that the slow variables equations well represent the main features of carbon dioxide vibrational dynamics.

The diversity of slow variables dynamics consists in wide variations in amplitude and period of intermode energy exchange that is completely determined by initial conditions, which include the phase difference and the partitioning of energy among modes. The only term of the potential energy that is linear in the symmetric stretch and quadratic in the bend mode coordinates is responsible for significant low frequency intermode energy exchange. The other potential energy terms superimpose high frequency variations on this main behavior.

The energy exchange spans from complete energy transfer back and forth between the symmetric stretch and bend to the absence of such an exchange depending on the initial conditions.

All these findings are important for understanding the process of carbon dioxide internal energy exchange with bath molecules. It is a conventional point of view that the major part of the vibrational energy flux to the translational and rotational degrees of freedom of surrounding molecules occurs via low frequency modes, i. e. the bending mode in the case of carbon dioxide. The resonance conditions enhance the deactivation of the symmetric stretch via the energy exchange with the bend.

Nevertheless, at a variety of initial conditions the symmetric stretch – bend energy exchange is not efficient. Additionally, the bend excitation lasts only a small portion of the energy exchange period (see Figs. 2 and 3). Thus, it is of great importance

to study the influence of the interaction with bath molecules on the process of energy redistribution among internal states of carbon dioxide molecules.

Acknowledgments

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